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- **VISION is the first trial of a MET inhibitor to provide data on health utilities (preference-based measures of HRQoL) in patients with METex14 skipping NSCLC**
- Overall HRQoL scores remained stable with tepotinib treatment, with no meaningful change in EORTC QLQ-C30 or EQ-5D-5L scores up to 84 weeks
- EORTC QLQ-C10D and EQ-5D utilities showed moderate-to-high functioning and quality of life during tepotinib therapy until progression
- The increase in EQ-5D utilities with tepotinib before IRC-assigned progression exceeds the previously reported minimally important difference in cancer of 0.08¹
- Utility with tepotinib did not vary by prior treatment status, or by adenocarcinoma or squamous histology
- EORTC QLQ-C10D and EQ-5D utilities from matched data collection points were highly correlated, suggesting similarities between both utility instruments

- Tepotinib is a highly selective, oral, once-daily MET inhibitor^{2,3} that has been approved in the US for treatment of advanced NSCLC harboring *MTEx14* skipping^{4,5}
- Approval was based on the Phase II VISION trial (**Figure 1**), in which tepotinib showed durable clinical activity and was well tolerated in patients with advanced *MTEx14* skipping NSCLCs^{6,7}
- PROs were evaluated as a secondary endpoint using EORTC QLQ-C30 and the EQ-5D-5L questionnaires, and showed maintenance of overall HRQoL during tepotinib treatment^{6,8}
- Results were scored from 0 to 100, where a change of ≥ 10 points from baseline was considered to be the minimal clinically important difference; higher scores indicated improvement on EORTC QLQ-C30 global health status and EQ-5D-5L SAS scores
- Health utilities are HRQoL metrics reflecting patient preferences for different health states, and are expressed on a scale from 0 (dead) to 1 (full health)⁹
- Utilities are widely used to inform cost-effectiveness analyses in Health Technology Assessment (HTA)⁹
- To complement the clinical findings of VISION, we used PRO data collected in the trial to evaluate utilities in tepotinib-treated patients with *MTEx14* skipping NSCLC (Cohort A; data cut-off: July 1, 2020)

Key inclusion criteria - Advanced NSCLC (EGFR/ALK wild-type, all histologies) - Liquid and/or tissue biopsy - MET alterations (central lab) 1L, 2L or 3L treatment setting - ECOG PS 0 or 1	Cohort A METex1.4 skipping	Tepotinib 500 mg once daily until PD, intolerable toxicity or withdrawal of consent	Selected endpoints Primary: - ORR by IRC (RECIST v1.1) Secondary: - PROs (EORTC QLQ-C30, EQ-5D-5L)
	Cohort B METamp		
	Cohort C METex1.4 skipping		

[illegible]

PROs (EORTC QLQ-C30, EQ-5D-5L)

ROs were collected at Day 1, then every 6 weeks for 9 months, and then every 12 weeks during treatment

Tepotinib

PD

EoT Safety follow-up

Weeks

post-treatment follow-up discontinued tepotinib due to PD

- PRO questionnaire responses were used to derive utilities using US weights:
 - EORTC QLQ-C10D utilities were derived from EORTC QLQ-C30 responses using US value sets¹⁰
 - For EQ-5D utilities, EQ-5D-5L data were mapped to EQ-5D-3L responses using a crosswalk algorithm,¹¹ and utilities were obtained using the value set for EQ-5D-3L weights for the US^{12,13}
- To account for dependencies within the data (i.e., correlated repeated measurements within patients) when evaluating mean change over time, utilities were analyzed using linear mixed modeling
 - The linear mixed model included a random intercept and fixed effects for baseline utility and progression status (i.e., pre- or post-progression health status, as assessed by IRC or INV):
 - Utility ~ 1 (patient ID) + baseline + progression status
- Exploratory analyses also evaluated the impact of prior treatment status, or adenocarcinoma or squamous histology
- Model fit was evaluated using the AIC and BIC

- At data cut-off (July 1, 2020), 152 patients were treated with tepotinib; 151 patients had confirmed *MEETx14* skipping and were analyzed for HRQoL, with a median follow-up for overall survival of 16.4 (range: 0.3 to 43.3) months
- Of 151 patients, half of the patients were male (52.3%), most were elderly (median age 73.0 years) with an ECOG PS of 1 (73.5%), half had a history of nicotine use (51.7%), and 54.3% had received prior therapy
- Questionnaire completion numbers were high (**Table S1**); accessible via QR code on the bottom right of the poster)

- Mean baseline scores of EORTC QLQ-C30 global health and EQ-5D-5L VAS showed moderate-to-high functioning and quality of life (54.3 [SD: 24.2] and 62 [SD: 20.4], respectively)
- Mean change from baseline in EORTC QLQ-C30 global health and EQ-5D-5L VAS scores demonstrated stability in patient quality of life over time (**Figures 3 and 4**)

- Of 151 patients analyzed for HRQoL, 983 and 907 observations were available for estimating EORTC QLU-C10D and EQ-5D utilities, respectively
- In linear mixed model analyses:
 - EORTC QLU-C10D health utilities were found to be significantly associated with baseline utility and progression status by IRC, but not with prior treatment status (**Table S2**; accessible via QR code on the bottom right of the poster)
 - EQ-5D utilities were found to be significantly associated with baseline utility and progression status by IRC, but not with prior treatment status or histology (**Table S3**; accessible via QR code on the bottom right of the poster)
- Therefore, separate utility values for baseline and pre- and post-progression health states were included in the analysis, irrespective of prior treatment for EORTC QLU-C10D health utilities, and irrespective of prior treatment or histologic subtype for EQ-5D utilities
- Estimated mean EORTC QLU-C10D utilities increased after tepotinib initiation, from 0.691 at baseline to 0.722 in the IRC-assessed progression-free period, and decreased after progression (0.671; **Figure 5**)
- Estimated mean EQ-5D utilities increased after tepotinib initiation, from 0.727 at baseline to 0.787 in the IRC-assessed progression-free period, and decreased after progression (0.733; **Figure 5**)
- Similar trends were seen when progression was based on INV assessment (**Figure 6**)

Time (weeks)	EORTC QLQ-C100 utilities [†] (Mean utility)	EORTC EQ-5D utilities [‡] (Mean utility)
0	0.691 (± 0.013)	0.727 (± 0.011)
4	0.722 (± 0.015)	0.787 (± 0.012)
8	0.671 (± 0.012)	0.733 (± 0.011)

Time (months)	Estimated EQ-V (Mean utility)	Estimated EQ-SD utilities (Mean utility)
0	0.694 (± 0.012)	0.694 (± 0.012)
6	0.785 (± 0.012)	0.727 (± 0.011)
12	0.726 (± 0.011)	0.669 (± 0.012)

- Utilities measured with each method (EORTC QLU-C10D and EQ-5D) were generally similar in all health states (**Figures 5 and 6**)
- Matched EORTC QLU-C10D and EQ-5D utilities derived from data collected at the same visit were generally highly correlated (Pearson's correlation coefficient: 0.681; **Figure 7**)

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